



CHALLENGE 17 / 20

THE EMAIL ATTACHMENT THAT OPENED THE DOOR

RETROACTIVE INITIAL ACCESS IDENTIFICATION

NATIONAL LEVEL HACKATHON • 2026

POWERED BY SYNTHOQUEST PVT LTD

Student Project Development Challenge • No organizer-provided evidence package.

1 | PROBLEM BACKGROUND

A law firm with 3,500 staff in 22 offices, advising on mergers and arbitrations for public-sector undertakings, found an adversary inside its network. The earliest evidence is a service created 71 days ago on a partner's workstation. The mail gateway keeps metadata for 30 days. Mailbox retention purged older mail, and the archive lost two months of index.

In the plausible window 4.1 million messages and 1.2 million attachments passed through, from clients, courts and opposing counsel. Entry may not have been email at all: VPN, a contractor and an exposed service are all possible. If it was email, the first message may have been harmless and a later link may have done the harm. The sender may be a trusted counterparty whose account was itself compromised.

Several staff reported strange attachments. All were later judged benign.

2 | PROBLEM STATEMENT

Develop an **Initial Access Vector Identification System** that, from downstream intrusion evidence and degraded mail and ingress metadata, identifies the most probable initial access vector, including whether it was email at all.

If it was email, the system must identify the message, thread, sender relationship, recipient and action that started the intrusion, reconstruct the multi-stage delivery chain, and state what it has ruled in and out.

Hard constraints

- The originating messages no longer exist and the candidate set runs to millions.
- The entry point may not be email, and the system must be able to say so.
- Delivery may be multi-stage, with a harmless first message.
- A trusted compromised sender must be told apart from legitimate correspondence.
- Teams must use benign stand-ins. No functional malware may be created or used.

No evidence package or dataset is provided. Teams decide what to build, what data they need, how to simulate the environment and how to validate the result. No tools or architecture are prescribed.

3 | MANDATORY FEATURES

Teams must develop a working software solution. The system must provide the following features. How each is achieved is for the team to decide.

01	Entry-Vector Discrimination	Ability to compare email against VPN, contractor and exposed-service entry.
02	Intrusion Start Estimation	Ability to estimate when and where the intrusion began.
03	Candidate Narrowing at Scale	Ability to narrow millions of candidates to a small ranked set.
04	Multi-Stage Delivery Reconstruction	Ability to rebuild a chain from a harmless first message to a harmful second stage.
05	Trusted-Sender Compromise Detection	Ability to detect a compromised counterparty.
06	Recipient and Action Identification	Ability to identify who did what to trigger entry.
07	Ruled-Out Reporting	Ability to state what has been excluded and why.
08	Calibrated Candidate Confidence	Ability to give measured confidence per candidate.

The system must clearly distinguish between:

IDENTIFIED	PROBABLE	POSSIBLE	RULED OUT	UNKNOWN
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An entry point that cannot be supported must not be presented as established.

Build, don't search. This is a development challenge. Teams must create the system and the controlled environment that proves it works. Submitting an analysis of a given dataset is not a solution.

4 | EXPECTED OUTPUT

Live demonstration. The team must show that its system can handle a complex controlled scenario created by the team. The demonstration must show:

01	Teams' own simulated mail environment with millions of messages and degraded retention
02	Several entry routes, including non-email
03	Estimate of intrusion start
04	Ranked entry candidates
05	Reconstruction of a multi-stage delivery
06	Exclusion statement
07	Calibrated confidence against ground truth

The team must be able to explain why the system reached each major conclusion.

Validation standard (every team)

- Build a scenario generator with known ground truth and evaluate on at least ten distinct scenarios, including ones with deliberately misleading evidence.
- At the demonstration, judges change scenario parameters on the spot. The system must adapt without being rebuilt.
- Include an adversary variant that actively tries to defeat the system, and report how the system held up.
- Stated confidence must match measured accuracy. Overconfident or unexplained conclusions lose marks.
- No real systems, real personal data or functional malware may be used.

Submission. Each team must submit:

01	Working Prototype	A functional implementation of the proposed solution.
02	Source Code	Complete source code required to reproduce the project.
03	Technical Architecture	Architecture and component description.
04	Validation Environment	A controlled environment created by the team to demonstrate the solution.
05	Technical Documentation	Implementation, design and testing documentation.
06	Forensic Investigation Report	A professional report generated from the demonstrated investigation.
07	Demonstration	Live technical demonstration to the judging panel.
08	Limitations	Known limitations and situations where the system cannot establish a reliable conclusion.